Numerical solution and validation of financial SDEs

Geometric Brownian motion and a non-affine stochastic-volatility model

Team F

MEMBER	STUDENT ID	ROLE
Kunze Li	*2363973*	*Project Manager*
Che Li	*2362072*	*Algorithm Implementation*
Zekun Wang	*2363718*	*Validation & Visualization*
Yixuan Liu	*2361140*	*Testing & Validation*
Yawen Wu	*2360749*	*Mathematical Theory*
Ruitong Huang	*2360294*	*Visualization & Report*

Xi'an Jiaotong-Liverpool University · October 2026

Two questions, answered against two kinds of reference

1. For geometric Brownian motion, whose exact solution is known, do coupled Monte Carlo experiments recover the theoretical strong and weak orders of Euler–Maruyama and Milstein once sampling uncertainty is quantified?
2. For a non-affine stochastic-volatility model without an exact transition law, does Euler–Maruyama refine at half order against nested numerical references, and can its weak payoff bias be resolved statistically?

WHY BOTH. Strong error is pathwise, weak error concerns expectations; their rates differ.

GOAL. A trend counts only where it exceeds the Monte Carlo uncertainty.

In Direction 4, stability regions and adaptive steps are replaced by the exact–EM–Milstein comparison, coupled strong and weak convergence, Monte Carlo confidence intervals and nested-grid validation; we add an analytic positivity analysis.

GBM: the exact solution and two one-step schemes

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t, \qquad S_T = S_0 \exp\!\left[(\mu - \tfrac{1}{2}\sigma^2)T + \sigma W_T\right],$$

$$S_{n+1}^{\mathrm{EM}} = S_n(1 + \mu h + \sigma \Delta W_n), \qquad S_{n+1}^{\mathrm{Mil}} = S_n\left[1 + \mu h + \sigma \Delta W_n + \tfrac{1}{2}\sigma^2\left((\Delta W_n)^2 - h\right)\right].$$

Errors are measured on the same Brownian path,

$$e_s(h) = \mathbb{E}\left|S_N^{(h)} - S_T\right|, \qquad e_w(h) = \left|\mathbb{E}S_N^{(h)} - \mathbb{E}S_T\right|,$$

and theory predicts strong order $\frac{1}{2}$ for Euler–Maruyama, 1 for Milstein, and weak order 1 for both.

Positivity. The Euler–Maruyama factor $1 + \mu h + \sigma\sqrt{h}\,Z$ can be negative for every h, the Milstein factor only if $(\sigma^2 - 2\mu)h > 1$; here $\sigma^2 - 2\mu = -0.06$ and $p_{\mathrm{EM}}(1/2) \approx 2.12 \times 10^{-13}$.

$S_0 = 100$, $\mu = 0.05$, $\sigma = 0.20$, $T = 1$; $M = 2 \times 10^6$ paths, seed 32104; $N = 2, 4, \ldots, 512$; coarse increments are block sums of one 512-step master path.

A non-affine volatility model with non-commutative noise

$$dX_t = \left(\mu - \tfrac{1}{2} g(Y_t)^2\right) dt + g(Y_t)\, dW_t^{(1)}, \qquad dY_t = \kappa(\theta - Y_t)\, dt + \xi \sqrt{1 + Y_t^2}\, dW_t^{(2)},$$

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \qquad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho\, dt, \qquad S_t = e^{X_t}.$$

Proposition *(non-commutativity).* *With $q(y) = \xi\sqrt{1+y^2}$, the diffusion columns b_1, b_2 of (X, Y) for independent drivers $B^{(1)} = W^{(1)}$, $B^{(2)}$ satisfy*

$$L^2 b_1 - L^1 b_2 = \left(\sqrt{1-\rho^2}\, q(Y)\, g'(Y),\, 0\right)^\top \neq 0.$$

Strong order one would in general need Lévy areas (commutator X-component 0.0433 at $Y = \theta$); we use log-Euler–Maruyama, for which $S_{n+1} = e^{X_{n+1}} > 0$.

Model: $\mu = 0.05$, $\kappa = 2$, $\theta = -0.2$, $\xi = 0.6$, $\sigma_{\min} = 0.10$, $\sigma_{\max} = 0.50$.
Initial values, horizon and correlation: $S_0 = 100$, $Y_0 = 0$, $T = 1$, $\rho = -0.7$; payoff $(S_T - 100)^+$.

Two exact identities serve as oracles for the code

Proposition *(GBM mean bias). For Euler–Maruyama and Milstein,* $\mathbb{E}S_N = S_0(1+\mu h)^N$*, hence*

$$\delta(h) = S_0\left[(1+\mu h)^{T/h} - e^{\mu T}\right] = -\tfrac{1}{2} S_0 e^{\mu T}\mu^2 T\, h + O(h^2).$$

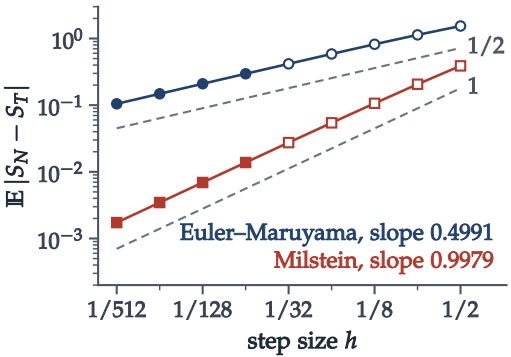
For our parameters $\delta(1/2) = -0.0646$; with $\mu = 0$ the bias would vanish and $\varphi(s) = s$ could not reveal a weak rate.

Proposition *(log-EM preserves the mean). For the non-affine log-Euler–Maruyama scheme and every* $h = T/N$*,* $\mathbb{E}S_N = S_0 e^{\mu T} = \mathbb{E}S_T$*.*

Remark. The identity payoff therefore carries no discretisation bias in the non-affine model; its weak error must be measured with a nonlinear payoff, here the call $(S_T - 100)^+$.

Second identity as a code oracle: the paired difference $S_N^{(h)} - S_N^{(h_{\text{ref}})}$ must be zero within error.

On GBM the strong orders one half and one are reproduced



Strong error against the exact coupled path, $M = 2 \times 10^6$. Filled points ($N \geq 64$) fitted; 95% bars within markers; dashed: slopes $\frac{1}{2}$ and 1.

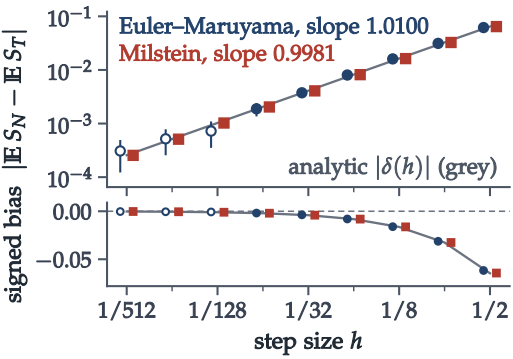
Fitted slopes, ± one jackknife SE:

Euler–Maruyama	0.4991 ± 0.0004
Milstein	0.9979 ± 0.0002

Local orders rise monotonically to 0.500 and 0.999; at $h = 1/512$ Milstein is about $60\times$ more accurate.

The coarse-grid deficit is pre-asymptotic curvature, so only $N \geq 64$ is fitted; the small gap to $\frac{1}{2}$ and 1 is finite-grid curvature, which the SE does not measure.

The weak bias of GBM is first order, as the analytic bias predicts



Top: absolute bias and analytic $|\delta(h)|$; bottom: signed bias with 95% intervals. Filled points fitted; markers offset horizontally for visibility.

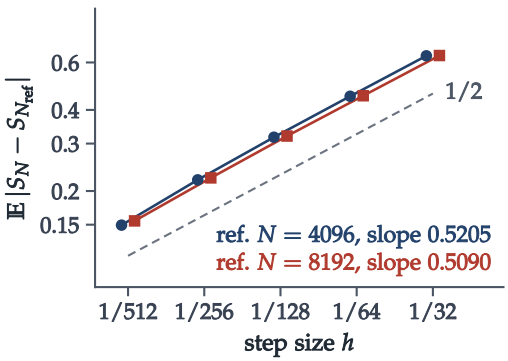
Weak slopes:

Euler–Maruyama	1.0100 ± 0.0440
	$N = 2$–64
Milstein	0.9981 ± 0.0004
	$N = 2$–512

All 18 signed intervals contain $\delta(h)$.

At $N = 256$, moments and all ten quantile intervals agree with the exact law within Monte Carlo error.

Log-EM shows an empirical half order against two references



Coupled discrepancy to Euler–Maruyama references, $M = 5 \times 10^4$. All points fitted, offset horizontally; 95% bars within markers; dashed: slope $\frac{1}{2}$.

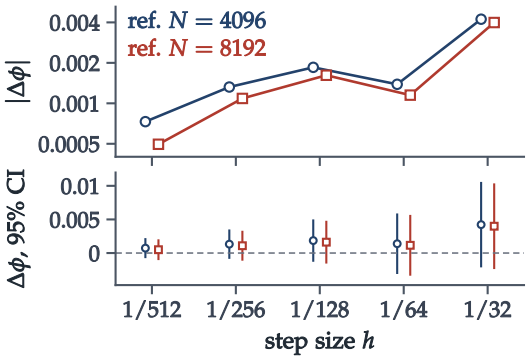
Slopes on $N = 32$–512:

ref. 4096 0.5205 ± 0.0019
ref. 8192 0.5090 ± 0.0018

Local orders rise at the finest grids (to 0.558 against 4096, 0.530 against 8192): a reference only $8\times$ finer shares part of the coarse error, so the finer reference is closer to $\frac{1}{2}$.

The references differ by 0.04022 ± 0.00015 (25.96% of the value at $N = 512$): these are discrepancies, not exact errors.

With 5×10^4 paths the non-affine payoff bias is not resolved



Paired call-payoff differences against both references; top: absolute, bottom: signed with 95% intervals. All points open (none fitted), offset horizontally.

All ten intervals contain zero (paired, 95%; SE about 0.003 at $N = 32$ and 0.0008 at $N = 512$).

No weak slope is reported: no grid passes the resolution rule in this run.

Common random numbers make the paired SE small, but for $N \geq 32$ the bias is smaller still; a slope fitted here would measure sampling error, not the scheme. E2 resolves the bias on coarser grids.

With 4×10^5 paths on coarse grids the payoff bias is resolved

N	payoff difference, ref. 512	payoff difference, ref. 1024	mean difference, ref. 1024
2	0.0719 ± 0.0046	0.0724 ± 0.0046	0.0024 ± 0.0053
4	0.0402 ± 0.0033	0.0406 ± 0.0033	0.0014 ± 0.0038
8	0.0174 ± 0.0023	0.0178 ± 0.0023	-0.0009 ± 0.0027
16	0.0090 ± 0.0016	0.0094 ± 0.0016	0.0005 ± 0.0019
32	0.0040 ± 0.0011°	0.0045 ± 0.0011	0.0007 ± 0.0013
64	0.00196 ± 0.00077°	0.00241 ± 0.00079°	0.00071 ± 0.00092

The weak slopes are 1.0209 ± 0.0685 (ref. 512, $N = 2$–16) and 1.0127 ± 0.0829 (ref. 1024, $N = 2$–32). The mean difference is zero within error, as the identity requires.

$M = 4 \times 10^5$; $\pm$ one SE; ° not resolved by the fit rule; grid set fixed after a first run with $N = 1$.

The strong rates disappear when the coupling is removed

	coupled		uncoupled	
N	Euler–Maruyama	Milstein	Euler–Maruyama	Milstein
8	0.8179	0.10665	23.573	23.544
32	0.41633	0.027503	23.606	23.599
128	0.20912	0.006937	23.615	23.613
512	0.10492	0.0017351	23.617	23.616
slope	0.4941	0.9906	−0.0004	−0.0007

Without coupling the error is the natural spread $\mathbb{E}|S_T - S'_T| \approx 23.62$ (E1), 29× to 225× the coupled Euler–Maruyama error. Replacing the block sum by a strided sum leaves the GBM slopes unchanged (0.501, 0.993) but lowers the non-affine slope from 0.519 to 0.028 (E1b).

E1: GBM strong error, $M = 2 \times 10^5$, independent seed stream. E1b: $M = 10^4$.

Cost at a matched strong-error target of 0.11

Method	N	strong error	upper 95%	work MN	solver time
Euler–Maruyama	512	0.10473	0.10485	1,024,000,000	5.18 ms
Milstein	8	0.10658	0.10672	16,000,000	0.199 ms

Milstein reaches the target with 64× fewer path updates and about 26× less wall-clock time; the second ratio is smaller because one Milstein step costs 3.02 ($N = 8$) to 2.55 ($N = 512$) times an Euler–Maruyama step.

The target 0.11 is about the smallest error Euler–Maruyama attains on the tested grids ($N \leq 512$). Random-number generation, not counted, would also favour Milstein.

GBM, $M = 2 \times 10^6$ for the errors. Time: solver loop only, 10,000 paths, median of 7 repeats, one machine. The 158× of our first version was a row-major storage artefact, now corrected.

Limitations

- The non-affine errors are discrepancies against Euler–Maruyama references, not exact errors; the reference gap is 25.96% of the finest discrepancy.
- The E2 weak slopes rest on coarse grids; the grid set was chosen after a first run.
- The call payoff is not smooth and e^X is not globally Lipschitz, so the non-affine rates are empirical.
- One parameter set; positivity was analysed only analytically; no Milstein scheme with Lévy areas.
- Slope errors are conditional jackknife SEs; timings come from one machine.

AI use (report, Appendix A). Codex: numerical library and tests, review-driven revisions (resolved weak grids, jackknife SE); on 10 Oct. (GPT-6 Astra, Ultra) the weak-grid refactor, independently re-checked. ChatGPT (GPT-6, Pro: review; GPT-6.1 Sol: language polishing). Grok Bot: LaTeX typesetting, report restructuring and code packaging; drafted the supplementary code and two derivations.

Summary of the observed rates

Study	Method	theory	slope ± SE	fitted N
GBM strong	Euler–Maruyama	1/2	0.4991 ± 0.0004	64–512
GBM strong	Milstein	1	0.9979 ± 0.0002	64–512
GBM weak	Euler–Maruyama	1	1.0100 ± 0.0440	2–64
GBM weak	Milstein	1	0.9981 ± 0.0004	2–512
Non-affine strong	ref. 4096 / 8192	(1/2)	0.5205 ± 0.0019 / 0.5090 ± 0.0018	32–512
Non-affine weak, E2	ref. 512 / 1024	(1)	1.0209 ± 0.0685 / 1.0127 ± 0.0829	2–16 / 2–32

On GBM the theory is reproduced against the exact solution; for the non-affine model the rates are empirical and depend on numerical references. Coupling is essential. Next steps: a third reference, multilevel Monte Carlo, and Milstein with simulated Lévy areas.

Theory in parentheses is expected, not proved, for the price statistic. *Thank you; questions are welcome.*